

Classification of Model Variables

Overview: We organized model variables into eight categories to reflect their clinical provenance and analytic roles across the perioperative pathway. The structure mirrors the style used for classifying clinical application modalities (title, succinct definitions, and key inclusion features). Each category is presented with scope, inclusion/exclusion rules, sub-categories, and representative examples to facilitate consistent variable mapping and downstream analyses.

Category 1 — Patient Baseline Characteristics

Scope: Fixed attributes, long-term health status, and lifestyle at admission/pre-procedure, including demographics, lifestyle, functional status, and chronic comorbidities.

Inclusion: Demographics (Age, Sex, BMI); Lifestyle (Smoking, Alcohol use); Functional status (ADL, Frailty index [mFI-5], Performance status); Comorbidities (COPD, Diabetes, Hypertension, CVD, CKD, Cirrhosis, Stroke).

Exclusion: Tumor staging (classified under Imaging/Tumor variables); Risk scores (classified under Risk Scores & Derived Features).

Sub-categories: Demographics; Lifestyle & Exposures; Functional status & Frailty; Chronic comorbidities.

Examples: Age ≥ 70 ; BMI; Current smoker; Pack-years; Alcohol > 2 drinks/day; ADL partially dependent; mFI-5; COPD; Diabetes; CKD; OSA; Stroke.

Category 2 — Preoperative Clinical & Laboratory Assessment

Scope: Preoperative organ function, physiologic status, nutrition, inflammatory/immune status, and routine laboratories.

Inclusion: Pulmonary: FEV1, FVC, DLCO; Cardiovascular: LVEF, NYHA class; Physiology/Gases: PaO₂/FiO₂, SpO₂, Respiratory rate; Nutrition: Albumin, Prealbumin; Routine labs: CBC, Chemistry, Coagulation; Inflammation/Immune: CRP, IL-6, NLR, TLR4, PNI.

Exclusion: Imaging-based tumor measures; Postoperative laboratories (classified under Early Postoperative Measures).

Sub-categories: Pulmonary function; Cardiovascular function; Nutritional biomarkers; Inflammatory/immune biomarkers; Routine laboratories.

Examples: FEV1 (% predicted); DLCO; NYHA $\geq$ III; SpO₂ $\leq$ 95%; Albumin < 35 g/L; WBC; Platelets; Sodium; INR; CRP; Procalcitonin; RDW; BUN/Creatinine.

Category 3 — Imaging and Tumor-related Characteristics

Scope: Structural findings from CT/MRI/PET and tumor diagnostics/staging plus radiotherapy parameters.

Inclusion: CT findings (Nodule, GGO %, Consolidation, Pleural effusion); Tumor staging (TNM); Tumor burden (Size, Volume); Radiotherapy planning (PTV volume, V20, V5, MLD [Gy]).

Exclusion: Non-imaging physiologic data; Intraoperative or real-time imaging.

Sub-categories: Structural imaging findings; Tumor staging; Radiotherapy planning.

Examples: GGO proportion; UIP pattern; Lymph-node short axis; Tumor volume; TNM IIIA; PTV 300 cc; V20 28%; MLD 5.5 Gy.

Category 4 — Surgical and Anesthetic Characteristics

Scope: Operative timing, approach, complexity, anesthesia modality, and airway/ventilation strategy.

Inclusion: Surgical type: Cardiac, Thoracic, Upper abdominal, VATS; Extent/complexity: Incision size, Number of segments resected; Urgency: Elective, Urgent, Emergency; Anesthesia: General, Regional, Balanced; Airway: Intubation, Supraglottic device; Pre-incision ventilation: PEEP, FiO₂, One-lung ventilation.

Exclusion: Intraoperative physiologic monitoring (classified under Intraoperative Course and Monitoring); Postoperative ventilation duration (classified under Early Postoperative Measures).

Sub-categories: Surgical type & complexity; Anesthesia method; Airway management; Pre-incision ventilation settings.

Examples: Operation time; VATS lobectomy; Open thoracotomy; Emergency surgery; General anesthesia; OLV; PEEP 5 cmH₂O; FiO₂ 0.8.

Category 5 — Intraoperative Course and Monitoring

Scope: Real-time intraoperative physiology, ventilation parameters, extracorporeal circulation, transfusion, and pharmacologic interventions.

Inclusion: Vitals: BP, HR, Temperature, ETCO₂; Ventilator: Tidal volume, Pressure control, PEEP, FiO₂; Hemodynamics: CVP, Arterial line, Vasoactive drugs; Events: Blood loss, Transfusion, Arrhythmia, Desaturation; Extracorporeal circulation: CPB time, Cross-clamp time.

Exclusion: Surgical type itself (Category 4); ICU/postoperative indicators (Category 6).

Sub-categories: Physiological monitoring; Ventilation monitoring; Hemodynamic support; Procedural events.

Examples: VT 6 mL/kg PBW; PEEP 8; Norepinephrine 0.08 µg/kg/min; Blood loss 600 mL; PRBC 2 units; CPB 125 min; Arrhythmia event.

Category 6 — Early Postoperative Measures and Support

Scope: Physiologic, laboratory, and supportive care parameters within 0–72 h post-surgery to assess early recovery and PPCs.

Inclusion: Laboratories: Lactate, Hb, WBC, INR, AST/ALT; Respiratory support: Duration of mechanical ventilation, Re-intubation; Circulatory/organ support: Vasoactive requirements; ICU-related: ICU admission, ICU hours; Thoracic specifics: Chest tube duration, Air leak; Analgesia/Sedation.

Exclusion: Long-term outcomes (e.g., 30-day mortality, LOS—consider as outcomes).

Sub-categories: Postoperative labs; Respiratory support; Hemodynamic/organ support; Surgical recovery indicators.

Examples: Postop lactate 3.0 mmol/L; Mechanical ventilation 12 h; ICU admission; Chest tube 48 h; RASS –1; Re-intubation.

Category 7 — Risk Scores and Derived Features

Scope: Composite variables derived from multi-dimensional data, including risk scores, administrative codes, engineered features, NLP features, and genetic markers.

Inclusion: Clinical risk scores: ASA, ARISCAT, EuroSCORE II, Charlson, PNI; Administrative coding: ICD-10, CPT, Phecodes; Derived variables: Ratios/Composite indices (NLR/LMR/CAR); NLP features: TF-IDF, Embeddings; Genetic/Molecular: FLT1, VEGFC, NRAS variants; Pediatric-specific items (as a sub-set).

Exclusion: Raw laboratory indicators (Category 2); Raw imaging features (Category 3).

Sub-categories: Clinical risk scores; Administrative coding; Derived quantitative features; NLP-based features; Genetic/Molecular markers; Pediatric-specific features.

Examples: ASA III; ARISCAT 45; Charlson ≥ 3 ; ICD-10: J95.1; NLR 4.3; TF-IDF features; FLT1 mutation.

Category 8 — Unclassified or Pending Information

Scope: Variables not yet mapped to the above seven categories, including institutional/system factors, ERAS measures, drug exposure details, emergency/trauma specifics, and other special items; to be reconciled later as needed.

Inclusion: Institutional/System: Region; Hospital bedsize; Payer (Medicare/Medicaid); Country income $\times$ SARS-CoV-2 risk. ERAS/Perioperative care: Implementation of ERAS. Drug exposures: Vancomycin trough; MED_penicillins; MED_anticoagulants; Remifentanyl; Alfentanil; Atracurium; Rocuronium; Neostigmine. Emergency/Trauma: TIE (time-to-ED after injury); GCS; Midline shift. Other: Work RVU; Admission source; Developmental delay.

Exclusion: Variables already assigned to Categories 1–7.

Sub-categories: Institutional & System Factors; ERAS & Perioperative Care; Drug Exposure Details; Trauma/Emergency Specifics; Miscellaneous Special Variables.

Examples: South region; Medium bedsize; Medicare/Medicaid; Implementation of ERAS; Vancomycin trough; Remifentanyl; TIE; GCS; Work RVU.